

6.5 Graphing Two-Variable Linear Inequalities

Lesson Introduction

 Benchmark	MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.		 Learning Targets	- I can find the solution set of a linear inequality where one variable has a coefficient of 0. - I can graph the solution set of a linear inequality given in standard form or point-slope form on a coordinate plane.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.912.AR.3.10 Given a mathematical or real-world context, graph the solution set to a two-variable, quadratic inequality.	
 Essential Questions	- How can you find the solution set of a linear inequality where one variable has a coefficient of 0? - How do you graph the solution set of a linear inequality on a coordinate plane?			
 Lesson Narrative	In this lesson, students continue learning about two-variable inequalities and graphing the solution set. Students will collaborate to explore graphs of solution sets to two-variable, linear inequalities in slope-intercept form. They will then be introduced to graphing solution sets to two-variable, linear inequalities in standard and point-slope forms. Lastly, they will collaborate to explore graphs of the solution sets to inequalities with vertical and horizontal boundary lines.			
 Component Summary	6.5.1 Warm-Up (~5 min)	Inequality graph Exploration (SE p. 318)	6.5.3 Guided Instruction (10-15 min)	Introduction to graphing two-variable, linear inequalities in standard and point-slope form (SE p. 320)
	6.5.2 Collaboration (10-15 min)	Explore graphs of two-variable, linear inequalities in slope-intercept form (SE p. 318)	6.5.4 Collaboration (10 min)	Explore inequalities with vertical and horizontal boundary lines (SE p. 322)
	6.5.5 Wrap-Up (~5 min)	Ticket Out the Door graphing two-variable, linear inequalities (SE p. 323)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Test points <u>Additional Terms:</u> N/A		N/A	N/A

 Teacher Tips	Common Misconceptions - Students may confuse when to use the dotted or solid line when graphing the inequality. - Students may shade in the incorrect region when graphing an inequality, particularly when determining which region to shade for inequalities in standard or point-slope form.	Things to Consider Each activity showcases a different form that the inequality is presented in. Go through each activity on your own prior to facilitating this lesson to look for potential misconceptions which each form of linear inequalities could present.
	Literacy and Language Routines Think-Pair-Share In 6.5.2 and 6.5.4 Collaborations, students will have individual think time to process and respond to the task, and then work with a partner to discuss their process and come to a consensus on their answers.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure The content from this lesson is an extension of what students have learned about solving and graphing equations and solving inequalities. Through this lesson, students will relate previously learned concepts to new concepts as they develop and deepen their understanding.
	 Differentiate Instruction Carefully consider your student groupings/pairings for the Collaboration activities. For example, in 6.5.2 Collaboration, students will work together with inequalities written in slope-intercept form. If students can master graphing inequalities in this form, it can serve as a base for understanding how to graph the solutions to inequalities given in the other forms. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

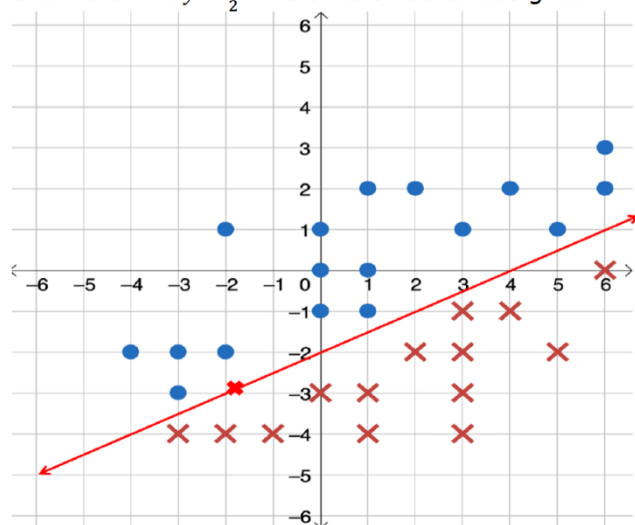
6.5.1 Warm-Up: Separating the X's and O's

Component Overview: Students will graph a given line to separate the X's and O's and answer three questions related to it.



6.5.1 Warm-Up: Separating the X's and O's

1. Draw the line $y = \frac{1}{2}x - 2$ on the coordinate grid.



- a. Which inequalities will include all of the circles?

$y \geq \frac{1}{2}x - 2$	$y \leq \frac{1}{2}x - 2$	$y > \frac{1}{2}x - 2$	$y < \frac{1}{2}x - 2$
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- b. Which inequalities will include all of the X's?

$y \geq \frac{1}{2}x - 2$	$y \leq \frac{1}{2}x - 2$	$y > \frac{1}{2}x - 2$	$y < \frac{1}{2}x - 2$
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- c. Draw an X on the graph so that only one of the inequalities will be true for all of the X's.

Answer on graph above.



This activity can provide insight into student retention and mastery on graphing two-variable, linear equations, and inequalities. For example, if students can graph question 1, but get parts a and b incorrect, they understand how to graph but may misunderstand the inequality component.

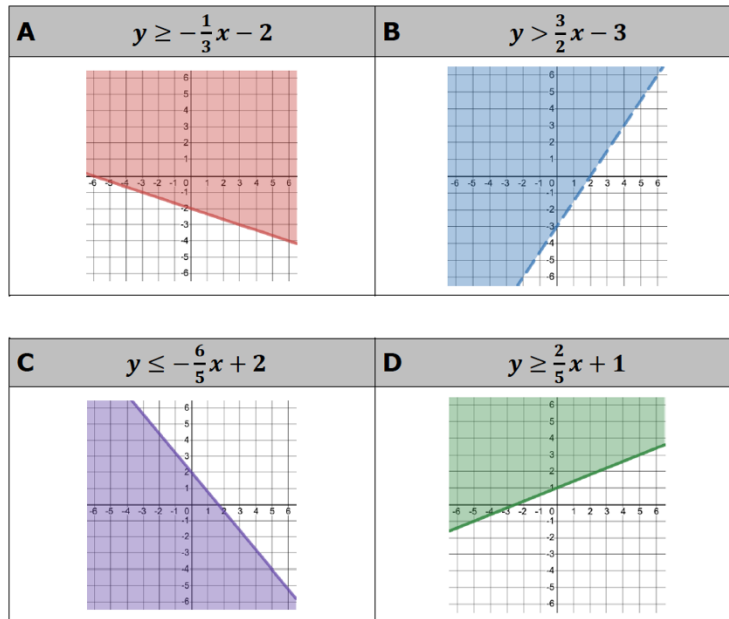
6.5.2 Collaboration: Two-Variable Linear Inequalities in Slope-Intercept Form

Component Overview: Students will collaborate to explore the solutions to two-variable, linear inequalities in slope-intercept form using three multipart questions. For this activity, students will need a partner or small group.



6.5.2 Collaboration: Two-Variable Linear Inequalities in Slope-Intercept Form

1. Four inequalities and their graphs are shown.



Complete the statements with the letter that corresponds to the inequality described.

- Inequality **B** is the only inequality where the points on the boundary line are not included in the solution set of the inequality.
- Inequality **C** is the only inequality where the points in the region below the boundary line are included in the solution set of the inequality.



For question 1, students will use what they know based on the inequality and its graph to answer parts a and b. For students who struggle getting started, consider prompting students to write down everything they know about the graphs, and use that to assist them with answering questions 1a and 1b.



- How can you determine that the other inequalities do NOT work to complete the statement in part a? Part b?
- What else do you notice or wonder about the graphs?

6.5.2 Collaboration: Two-Variable Linear Inequalities in Slope-Intercept Form (continued)

2. Consider the inequality, $y > -3x + 1$.

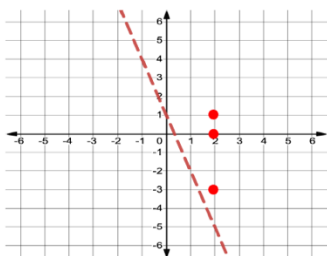
- a. Substitute $x = 2$ into the inequality $y > -3x + 1$ and evaluate.

$$y > -3(2) + 1$$

$$y > -5$$

- b. What values of y satisfy the inequality when $x = 2$?
 $y > -5$

The boundary line, $y = -3x + 1$, is graphed on the coordinate grid. The ordered pair $(2, -5)$ is a point on the boundary line.



- c. Discuss with your partner why the point $(2, -5)$ is not part of the solution set for the $y > -3x + 1$.
Answers may vary. The point $(2, -5)$ is not part of the solution set for $y > -3x + 1$ because if you substitute the ordered pair into the inequality, you would get $-5 > -5$, which is not true.

- d. Write three ordered pairs with x -coordinates of 2 that are solutions to the inequality $y > -3x + 1$.
Answers will vary.

$$(2, -3) \quad (2, 1) \quad (2, 0)$$

- e. Plot the three points you chose on the coordinate grid.
Answers will vary. Example shown above.



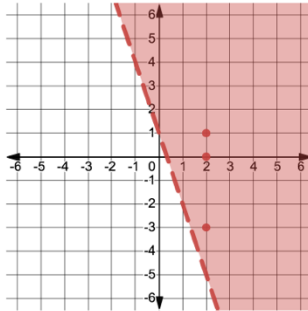
When you substitute the value of y that satisfies the inequality when $x = 2$, does it give you a value greater than -5 ?



Listen to student conversation for question 2c. Ensure that students understand that the point $(2, -5)$ is not part of the solution set. Some students may refer to the dotted line on the graph while others will use the inequality for their reasoning.

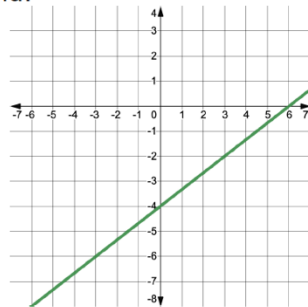
6.5.2 Collaboration: Two-Variable Linear Inequalities in Slope-Intercept Form (continued)

- f. Shade the solution set of the inequality $y > -3x + 1$.



3. Consider the inequality, $y \leq \frac{2}{3}x - 4$.

The boundary line, $y = \frac{2}{3}x - 4$, is graphed on the coordinate grid.



- a. Discuss with your partner why the boundary line is solid.

The boundary line is solid because the values on the line represent solutions.

- b. Select a value for x between -6 and 6 .

Answers will vary. 0.



Make a prediction from what you did in question 2. Which area of the graph do you think will be shaded for the inequality in question 3?

6.5.2 Collaboration: Two-Variable Linear Inequalities in Slope-Intercept Form (continued)

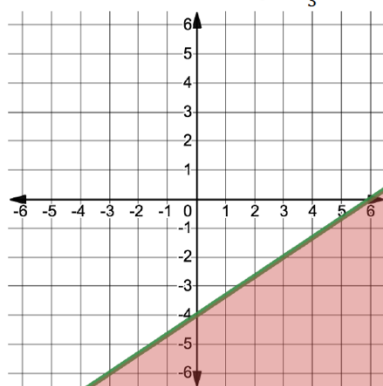
- c. Substitute your chosen value for x into the inequality $y \leq \frac{2}{3}x - 4$, and evaluate.

Answers will vary.

$$y \leq \frac{2}{3}(0) - 4$$

$$y \leq -4$$

- d. Based on your answer to Part C, shade the solution set of the inequality $y \leq \frac{2}{3}x - 4$.



- e. Discuss with your partner how you can determine whether the graph of an inequality in slope-intercept form is shaded above or below the boundary line. Summarize your discussion.

Answers will vary. You can determine whether the graph of an inequality in slope-intercept form is shaded above or below the boundary line by looking at the inequality symbol. If the inequality symbol is $>$ or $\geq$, then the graph is shaded above the boundary line. If the inequality symbol is $<$ or $\leq$, then the graph is shaded below the boundary line.



At the conclusion of this activity, consider having students share out their answers to question 3e. Engage students in a conversation that allows them to explain their answers and respond to their peers.

6.5.3 Guided Instruction: Two-Variable Linear Inequalities in Standard and Point-Slope Forms

Component Overview: Students will learn how to graph the solution set of two-variable, linear inequalities in standard and point-slope forms using three practice questions.



6.5.3 Guided Instruction: Two-Variable Linear Inequalities in Standard and Point-Slope Forms

Linear inequalities in two variables can be written in forms other than slope-intercept form.

1. The linear inequality $2x - 3y > -9$ is written in standard form.

- a. Complete the statement.

The boundary line of this inequality is

- ☐ dotted
☐ solid

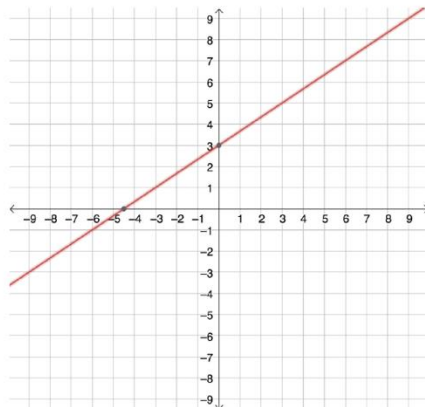
because the points on the boundary line are

- ☐ included in
☐ excluded from
- the solution set.

- b. Write the equation of the boundary line of this inequality in standard form.

$$2x - 3y = -9$$

- c. Graph the boundary line on the coordinate grid.



How do you know what type of boundary line should be used when graphing the solution to an inequality?



Consider using question 1 in a back-and-forth dialogue method of instruction to connect the concept to how to graph the solution set. Help reinforce the operational vocabulary for the form type and how they can connect this to what they learned in the prior component.

6.5.3 Guided Instruction: Two-Variable Linear Inequalities in Standard and Point-Slope Forms (continued)

- d. With your partner, discuss a conjecture about which side of the boundary line to shade to represent the solution set of the inequality.

Answers will vary.

- e. Select two test points, one on each side of the boundary line. Substitute the ordered pairs into the inequality to determine which ordered pair is in the solution set.

Answers will vary.

Coordinates of Test Point	(0,5)	(2,1)
Substitute the Coordinate Values Into the Inequality $2x - 3y > -9$	$2(0) - 3(5) > -9$ $-15 > -9$	$2(2) - 3(1) > -9$ $1 > -9$
Is it a solution to $2x - 3y > -9$?	☐ Yes ☒ No	☒ Yes ☐ No

- f. Before testing any points, Elias made the following conjecture about which region would be shaded.

"I think I need to shade above the boundary line because the inequality sign is $>$, which means greater than."

Based on your work in Part E, explain whether Elias' conjecture was correct.

Answers will vary. Elias was incorrect because the test point (2,1) was a solution, but the point (0,5) was not, so the graph is shaded below the boundary line.



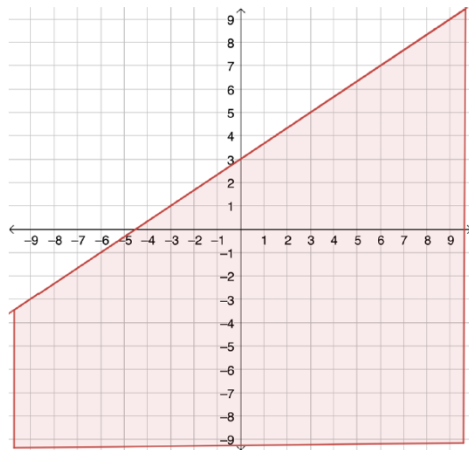
Release students to discuss question 1d with their partners and then call the students back together for a whole-group share-out of their responses to allow a chance for hearing from their peers.



Students may have the misconception that, like with slope-intercept form, the "greater than" symbol indicates that the graph should be shaded above the line. Keep curiosity here, as this misconception will soon be addressed.

6.5.3 Guided Instruction: Two-Variable Linear Inequalities in Standard and Point-Slope Forms (continued)

- g. Shade the region on the graph that represents the solution set.



When graphing a linear inequality in two variables, **test points** can be selected and substituted into the inequality to determine which region should be shaded to represent the solution set.

2. Explain how it could be possible to determine which region to shade using only one test point.

Answers will vary. Since in the graph of a linear inequality in two variables only one side of the boundary line is shaded, you can just use one test point. If that test point satisfies the inequality, you shade the side that includes the test point. If the test point does not satisfy the inequality, you shade the side that does not include the test point.



Add the language “test points” to any existing vocabulary strategies such as anchor chart, word wall, vocabulary flip chart, reference guide, or Frayer Model.

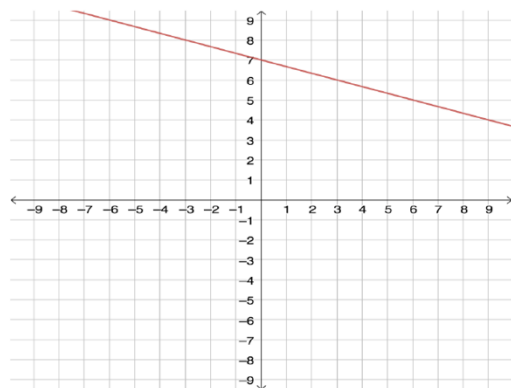


If pacing is a concern, consider converting question 2 into a whole-group discussion where students can respond in real time, instead of having to write in their book.

6.5.3 Guided Instruction: Two-Variable Linear Inequalities in Standard and Point-Slope Forms (continued)

3. The inequality $y - 8 \leq -\frac{1}{3}(x + 3)$ is given in point-slope form.

- a. Graph the boundary line of the inequality.



- b. Select one test point that is not on the boundary line. Substitute your selected point into the inequality $y - 8 \leq -\frac{1}{3}(x + 3)$ to determine whether it is in the solution set. Show your work.

Answers will vary based on the selected point.

- c. Shade the region to represent the complete solution set of the inequality $y - 8 \leq -\frac{1}{3}(x + 3)$.



For question 3, consider leveraging a Think-Pair-Share routine. Release students to try a, b, and c on their own, and then check their work with a partner to come to a consensus on an answer. Have students check their partner work with the partner work of another partner pair to confirm their answer before going over the question whole-group.

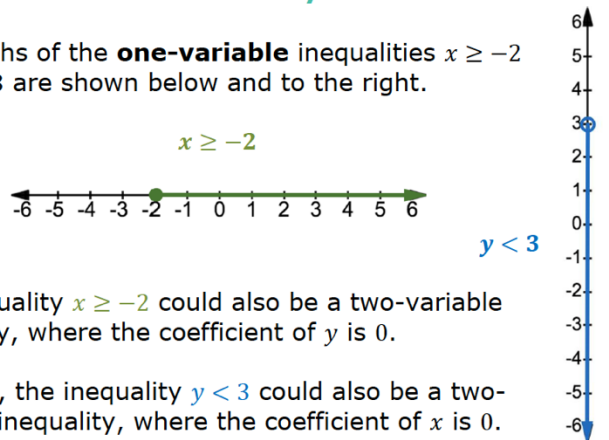
6.5.4 Collaboration: Inequalities With Vertical and Horizontal Boundary Lines

Component Overview: Students will collaborate to work with inequalities with vertical and horizontal boundary lines using six different inequalities. For this activity, students will need a partner or small group.



6.5.4 Collaboration: Inequalities With Vertical and Horizontal Boundary Lines

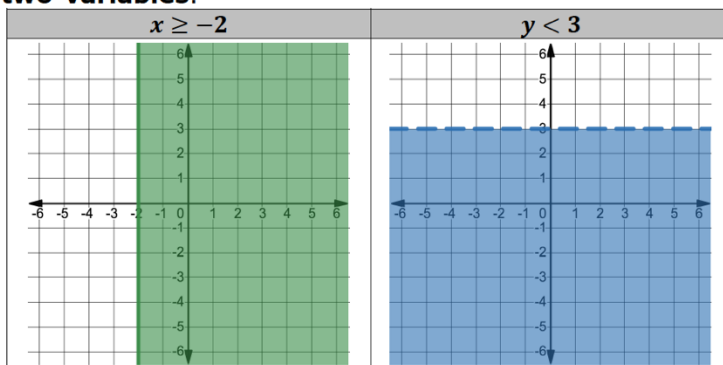
The graphs of the **one-variable** inequalities $x \geq -2$ and $y < 3$ are shown below and to the right.



The inequality $x \geq -2$ could also be a two-variable inequality, where the coefficient of y is 0.

Similarly, the inequality $y < 3$ could also be a two-variable inequality, where the coefficient of x is 0.

The inequalities $x \geq -2$ and $y < 3$ are shown graphed in **two-variables**.



- How do the two graphs in two variables relate to the corresponding graphs in one variable?

Answers will vary. The graphs in two variables look like the graphs in one variable just got repeated over and over.



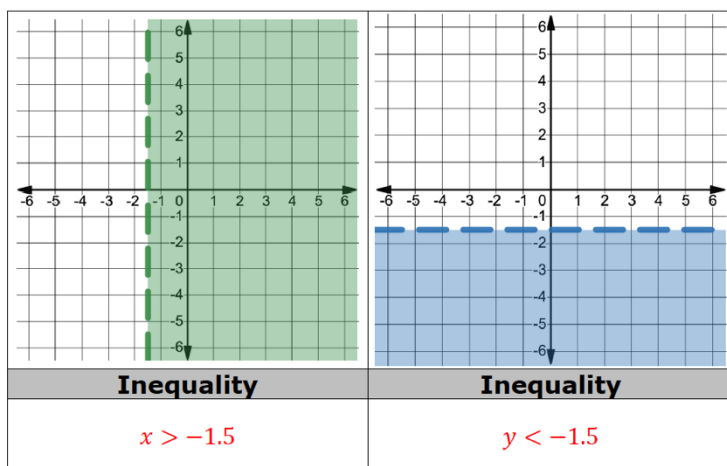
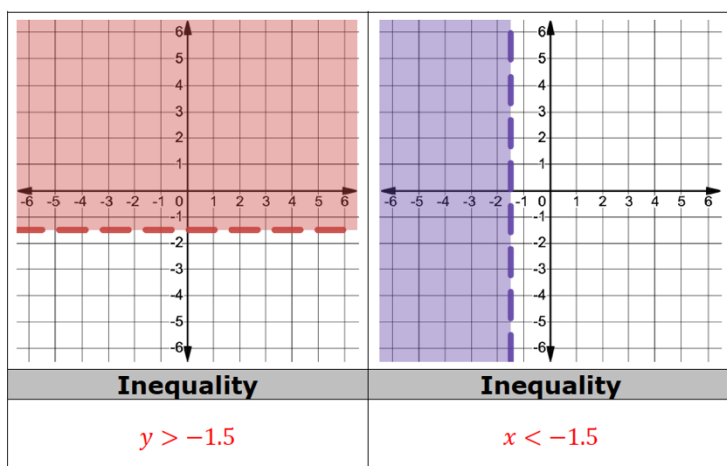
What similarities and differences do you see with these inequalities and equations of a similar type?

6.5.4 Collaboration: Inequalities With Vertical and Horizontal Boundary Lines (continued)

2. Four inequalities are given.

$$x > -1.5 \quad x < -1.5 \quad y > -1.5 \quad y < -1.5$$

Match each inequality to its graph in two variables.



How did you determine which region to shade for these types of inequalities?

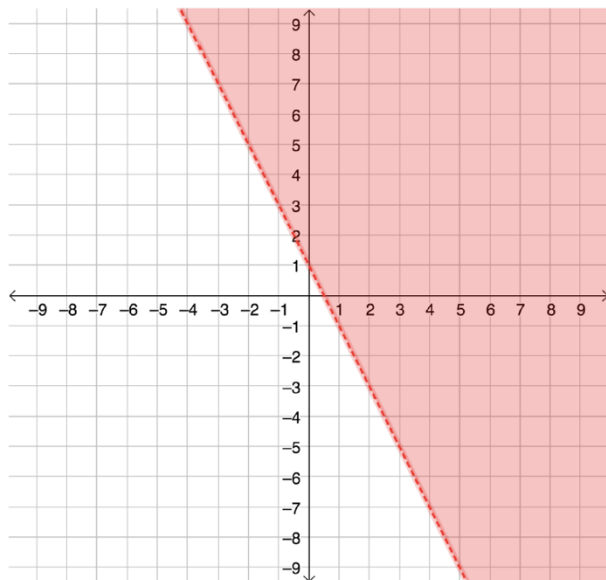
6.5.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door, graphing an inequality given in point-slope form.



6.5.5 Wrap-Up: Ticket Out the Door

- Graph the inequality $y + 5 > -2(x - 3)$.



- Explain how you determined which part of the graph to shade.

Answers will vary. I used the test point (2,1) and found that it was in the solution set of the inequality.



Student performance on this question can provide information about where students may have a misunderstanding in the process of graphing an inequality given in point-slope form. This information can be used to plan for differentiated support in future lessons.